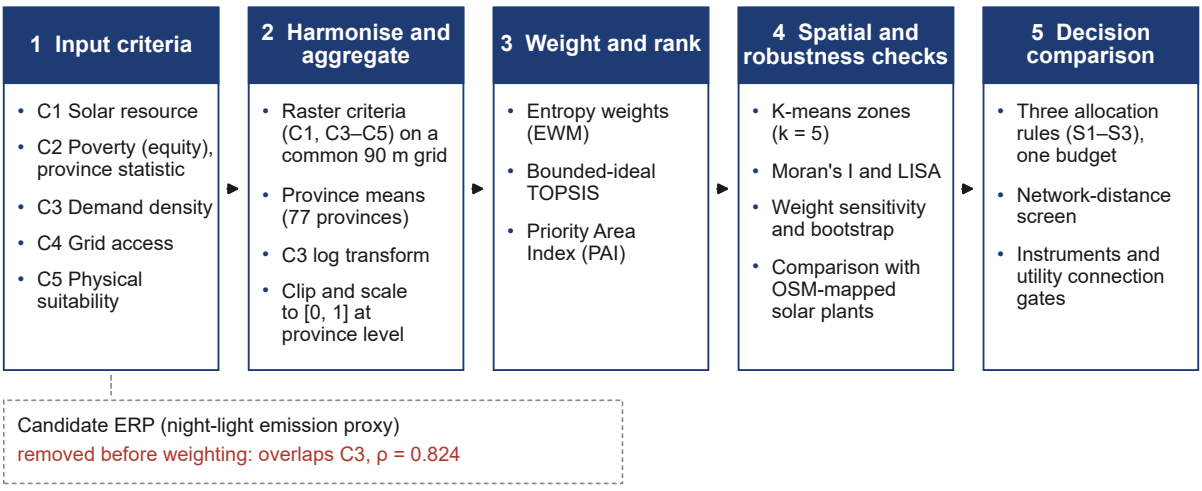
1 Input criteria
C1 Solar resource
C2 Poverty (equity), province statistic
C3 Demand density
C4 Grid access
C5 Physical suitability
2 Harmonise and aggregate
Raster criteria (C1, C3–C5) on a common 90 m grid
Province means (77 provinces)
C3 log transform
Clip and scale to [0, 1] at province level
3 Weight and rank
Entropy weights (EWM)
Bounded-ideal TOPSIS
Priority Area Index (PAI)
4 Spatial and robustness checks
K-means zones (k = 5)
Moran's I and LISA
Weight sensitivity and bootstrap
Comparison with OSM-mapped solar plants
5 Decision comparison
Three allocation rules (S1–S3), one budget
Network-distance screen
Instruments and utility connection gates
Candidate ERP (night-light emission proxy)
removed before weighting: overlaps C3, ρ = 0.824